

Consumer sleep-tracker validation against PSG, 2-stage sleep/wake: one dot per device in each study (primary analysis rows only)

Dashed green = candidate good zone (TST bias ± 15 min; LoA ± 60 min; sensitivity $\geq 92.5\%$; specificity $\geq 47.5\%$; accuracy $\geq 85\%$; $\kappa \geq 0.425$; PABAK ≥ 0.725 ; F1 ≥ 0.90). Whiskers = 95% CI where reported. LoA panel: dot = bias, bar = 95% LoA; dotted bar = LoA derived here as bias ± 1.96 SD (paper reported SD only). F1 (sleep) is derived, not reported: exact from epoch counts or wake-class F1 where published, otherwise estimated from sensitivity, specificity and PSG sleep prevalence (hollow markers). Sensitivity/specificity swapped to sleep-positive convention for Stucky 2021, Lee 2023 and Labie 2025. Diamonds = manufacturer white papers, not peer reviewed. '(median)' marks values that are medians of per-night values rather than means or pooled epochs (Apple 2025).

